NABIN KUMAR KC

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Education

Wichita State University **Jan 2025 – Dec 2026 (Expected)**
Master of Science in Computer Science *Wichita, Kansas*

Nitte Meenakshi Institute of Technology **Aug 2019 – July 2023**
Bachelor of Engineering in Computer Science Engineering *Bangalore, India*

Technical Skills

Programming: Python, SQL, C, JavaScript, HTML/CSS
Machine Learning: Scikit-learn, Random Forest, KNN, Logistic Regression, Linear Regression
Deep Learning: TensorFlow/Keras, CNNs, Image Classification, Data Augmentation
Computer Vision: OpenCV, MediaPipe, Image Preprocessing, Feature Detection
Data/Analytics: Excel, Pandas, NumPy, Matplotlib, Seaborn, EDA, Data Preprocessing, Feature Engineering
Tools: Git, GitHub, VS Code, PyCharm, Jupyter Notebook, Google Colab, Linux, Vercel
Cybersecurity: Wireshark, NFStream, Network Traffic Analysis

Experience

Wichita State University **Jan 2026 – Present**
Graduate Research Assistant (Part-time) *Wichita, Kansas*

- Developed an OCR pipeline that extracts key fields from scanned transcripts, application forms, and CVs using OpenCV, regex, and fuzzy matching, achieving 91% field-level accuracy.
- Evaluated OCR robustness on 360 simulated/scanned documents with phone-scan noise across three image-quality levels.
- Built and evaluated Decision Tree, Random Forest, and Logistic Regression models that turn extracted data into applicant profiles for Admit/Review/Deny decision support.
- Conducted benchmarking experiments comparing model performance and presented findings and recommendations to faculty advisors to guide the direction of the research.

YBI Foundation **Sep 2022 – Nov 2022**
Machine Learning with Python Intern *Remote*

- Performed exploratory data analysis and visualization on real-world datasets using Pandas, Matplotlib, and Seaborn to uncover trends and feature correlations.
- Preprocessed data by handling missing values, encoding categorical features, scaling, and applying PCA.
- Built Linear/Logistic Regression, KNN, and Random Forest models in Scikit-learn; evaluated regression with R^2 and classification with accuracy and confusion matrices.

The Sparks Foundation **Jun 2022 – Aug 2022**
Frontend Web Developer Intern *Remote*

- Designed and developed a responsive multi-page donation website (Home, Donate, Contact) for The Sparks Foundation using HTML, CSS, and JavaScript.
- Integrated the Razorpay payment gateway into the donation flow, enabling users to make secure online payments directly from the site.
- Implemented interactive animations using JavaScript and GSAP, along with a contact form to improve user engagement.

Projects

OCR-Based Admissions Document Extraction | *Python, OpenCV, OCR, Scikit-Learn* **Jan 2026 – Present**

- Built a 4-stage OCR pipeline (OpenCV preprocessing, adaptive thresholding, regex extraction, fuzzy matching).
- Developed an automated Admit/Review/Deny workflow combining OCR-extracted features with ML classifiers.

Hand Gesture Control System *(Published, Springer 2024)* | *Python, OpenCV, MediaPipe, Tkinter* **Jun 2023**

- Built a desktop app to control a computer with webcam hand gestures, using OpenCV and MediaPipe tracking.
- Mapped gestures to volume, brightness, mouse, and media controls via a Tkinter GUI packaged as an executable.

Malicious DoH Traffic Detection | *Python, TensorFlow, Scikit-Learn, NFStream*

- Detected malicious DNS-over-HTTPS (DoH) tunneling traffic using network flow features extracted with NFStream.
- Compared logistic regression, random forest, and CNN models; tested robustness to adversarial evasion.

Plant Disease Detection System | *Python, TensorFlow/Keras, CNN, OpenCV* **Dec 2025**

- Built a custom CNN for plant disease detection, with augmentation and normalization for robustness.
- Deployed the model in a web app where users upload leaf images for real-time disease predictions.